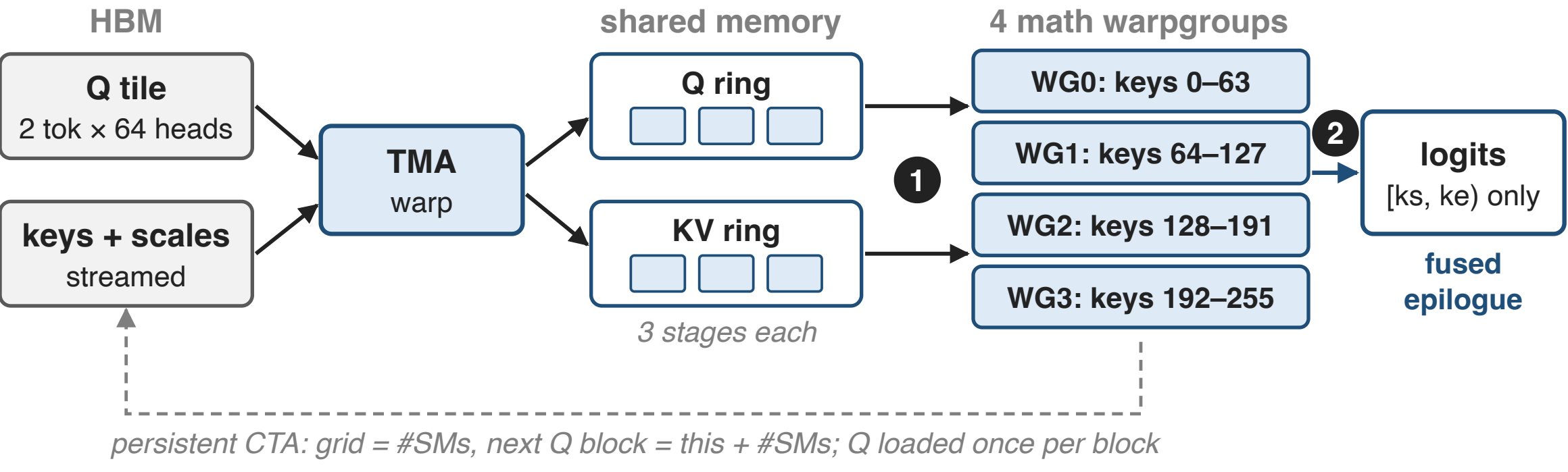
HBM
shared memory
4 math warpgroups
Q tile
2 tok × 64 heads
keys + scales
streamed
TMA
warp
Q ring
KV ring
3 stages each
1
WG0: keys 0–63
WG1: keys 64–127
WG2: keys 128–191
WG3: keys 192–255
2
logits
[ks, ke) only
fused
epilogue
persistent CTA: grid = #SMs, next Q block = this + #SMs; Q loaded once per block